SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A

Comparative Analysis

Code of Conduct:

I V. Thanusri(192472265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Thanusri

Rubrics for Evaluation

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive	Provides adequate comparison	Limited comparison with few aspects	Very minimal or no comparison	

		comparison across multiple aspects	with some depth			
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5

B 3.0 3.8

C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

2. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

3. Precision-Recall Trade-off Analysis

Two models produce:

Model TP FP FN

A 70 30 20

B 85 25 25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

4. F1-Score Comparison

Two models:

Model Precision Recall

A 0.8 0.6

B 0.75 0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

5. Optical Flow Consistency

Motion values:

Frame Method A Method B

1 4 6

2 5 8

3 6 9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

Answers

1. Depth Estimation Comparison (Stereo vs SfM)

Given:

Actual depths: A = 2 m, B = 3 m, C = 4 m.

Stereo estimates: A = 2.0 m, B = 3.0 m, C = 4.0 m.

SfM estimates: A = 2.5 m, B = 3.8 m, C = 5.0 m.

(a) Absolute Error

The absolute error is calculated using:

Absolute Error = |Estimated Depth − Actual Depth|

For Stereo Vision:

Object A: |2.0 − 2.0| = 0.0 m

Object B: |3.0 − 3.0| = 0.0 m

Object C: |4.0 − 4.0| = 0.0 m

Total absolute error = 0 + 0 + 0 = 0 m

Mean Absolute Error (MAE) = 0/3 = 0 m

For Structure from Motion (SfM):

Object A: |2.5 − 2.0| = 0.5 m

Object B: |3.8 − 3.0| = 0.8 m

Object C: |5.0 − 4.0| = 1.0 m

Total absolute error = 0.5 + 0.8 + 1.0 = 2.3 m

MAE = 2.3/3 = 0.767 m approximately.

(b) Comparison and Evaluation

Stereo Vision is more accurate because all three estimated depths are exactly equal to the actual depths. Its MAE is 0 m, whereas SfM has an MAE of approximately 0.767 m. SfM also consistently overestimates the actual depth. Therefore, based on the calculated errors, Stereo Vision is the better technique for this dataset.

2. Pattern Recognition Accuracy Comparison

Given:

Traditional system: 85 correct predictions out of 100 samples.
Deep Learning system: 92 correct predictions out of 100 samples.

(a) Accuracy

Accuracy is calculated using:
Accuracy = (Correct Predictions / Total Samples) × 100

Traditional system:
Accuracy = (85/100) × 100 = 85%

Deep Learning system:
Accuracy = (92/100) × 100 = 92%

(b) Comparison and Evaluation

The Traditional system achieves 85% accuracy, while the Deep Learning system achieves 92% accuracy. Therefore, Deep Learning is more accurate by:

92% − 85% = 7 percentage points.

In terms of correct predictions, the Deep Learning system correctly identifies 7 more samples out of every 100 samples in this dataset. Hence, the Deep Learning system performs better because it has the higher accuracy of 92%.

3. Precision-Recall Trade-off Analysis

Given:

Model A: TP = 70, FP = 30, FN = 20.

Model B: TP = 85, FP = 25, FN = 25.

(a) Precision and Recall

The formulas are:

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

Model A:

Precision = 70/(70 + 30) = 70/100 = 0.700 = 70.0%

Recall = 70/(70 + 20) = 70/90 = 0.778 = 77.8%

ModelB:

Precision = 85/(85 + 25) = 85/110 = 0.773 = 77.3%

Recall = 85/(85 + 25) = 85/110 = 0.773 = 77.3%

(b) Comparison and Evaluation

Model A has a precision of 70.0% and recall of 77.8%. It therefore detects a larger proportion of actual positive cases, but it also produces more false positives relative to its true positives.

Model B has both precision and recall equal to approximately 77.3%. The two measures are almost perfectly balanced. Model B also has higher precision than Model A while maintaining a recall close to that of Model A.

Therefore, Model B is the more balanced model because its precision and recall are equal at approximately 0.773. It provides a better trade-off between avoiding false positives and detecting actual positive cases.

4. F1-Score Comparison

Given:

Model A: Precision = 0.8, Recall = 0.6.

Model B: Precision = 0.75, Recall = 0.75.

(a) F1-Score

The F1-score is the harmonic mean of precision and recall:

F1 = 2 × (Precision × Recall)/(Precision + Recall)

For Model A:

F1 = 2 × (0.8 × 0.6)/(0.8 + 0.6)

= 0.96/1.4

= 0.686 approximately.

For Model B:

F1 = 2 × (0.75 × 0.75)/(0.75 + 0.75)

= 1.125/1.5

= 0.750.

(b) Comparison and Evaluation

Model A has a higher precision of 0.80, but its recall is only 0.60. The low recall reduces its F1-score to approximately 0.686.

Model B has equal precision and recall values of 0.75, resulting in an F1-score of 0.750. Since a higher F1-score indicates a better balance between precision and recall, Model B performs better overall.

Therefore, Model B is preferred because its F1-score of 0.750 is higher than Model A's 0.686, showing a more balanced classification performance.

5. Optical Flow Consistency

Given motion values:

Frame 1: Method A = 4, Method B = 6

Frame 2: Method A = 5, Method B = 8

Frame 3: Method A = 6, Method B = 9

(a) Average Motion

Average motion is calculated as:

Average = Sum of motion values / Number of frames

Method A:

Average = (4 + 5 + 6)/3

= 15/3

= 5.00

Method B:

Average = (6 + 8 + 9)/3

= 23/3

= 7.67 approximately.

Thus, Method B detects greater average motion.

(b) Comparison of Consistency and Stability

To evaluate consistency, we examine the variation in the motion values.

Method A:

Minimum = 4, Maximum = 6

Range = 6 − 4 = 2

Method B:

Minimum = 6, Maximum = 9

Range = 9 − 6 = 3

The approximate population standard deviations are:

Method A = 0.82

Method B = 1.25

A smaller standard deviation indicates less variation and therefore greater stability. Method A has both a smaller range (2) and smaller standard deviation (approximately 0.82) than Method B, whose range is 3 and standard deviation is approximately 1.25.

Therefore, Method A is more consistent and stable. Method B measures a higher average motion (7.67 compared with 5.00), but its values fluctuate more between frames.